Design of Semi-Physical Simulation Platform for Solar Array Drive Assembly

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Abstract. To solve the problem of the geostationary orbit (GEO) satellite perturbed by flexible solar wing for solar array drive assembly (SADA). A semi-physical platform centered on the electric load simulator is designed to accurately simulate the solar wing reaction moment with variable load characteristics. A load simulator dynamic model is developed, and an extended state observer is created by estimating the internal parameter errors of the system with external perturbations as composite disturbances. The effect of observation errors and disturbances on the system is eliminated by using the sliding mode control, and the system control quantity is calculated; finally, the stability of the system is proved by using the Liapunov method. The platform's response speed, loading accuracy, and error are analyzed under different signals, and the results show that the algorithm can simulate the desired solar wing reaction moment more accurately.

Keywords: Semi-Physical Simulation, Extended State Observer, Sliding Mode Controller, Torque Control.

1 Introduction

The energy source of the geostationary orbit satellite is mainly the flexible veneer-folded solar wing linked to the satellites [1]; the solar array is driven by solar array drive assembly (SADA) that adjusts the solar sail panels to the angle with the highest solar energy utilization. A flexible solar wing is characterized by large flexure, large inertia, rigid-flexible coupling, and dense modes, which lead to variable loads with large moments and high-frequency ringing during the driving process of SADA. Therefore, it is necessary to test the driving performance of SADA before the launch of the satellite [2].

The large inertia, large size, and modal-density characteristics of the flexible solar wing create difficulties for ground testing, mathematical modeling, and model correctness estimation. Li [3] and Zhou [4] used actual solar wing products for ground drive tests with good results but did not consider the effect of ground gravity. Yin [5] uses the air-floatation method to compensate for the gravity of the solar wing based on the above, but it is only applicable to solar wings with significant stiffness and has not been verified for solar panels with rigid-flexible coupling. Wang [6] uses inertia

rods and adjustment blocks to simulate biplane solar sail panels with different simulation characteristics, which realizes the function of adjustable inertia, but the simulation of rotational inertia of less than $10kg \cdot m^2$ could not be realized. Therefore, it is more difficult to realize the drive performance test of the SADA by conducting a ground-based all-physics test.

He [7] realizes the design and accuracy assessment of SADA utilizing a semi-physical platform, which can accurately reproduce the expected moments of the solar flap model, but mainly for the large-size flexible biplane sailplanes of the space station, and does not discuss the smaller-size rigid-flexible coupled solar wing. Liu [8] mainly discusses the overall structural design of the semi-physical platform and does not cover the control methods.

A dynamic model of the loading unit with an electric load simulator as the core is established according to the characteristics of low speed and large torque of the solar orientation device [9]. An expanded state observer [10] is designed for the internal errors and external nonlinear perturbations present in the modeling process, and a sliding mode control [11] law is used to reduce the effects of the perturbations. An accurate desired load moment [12] is simulated for the drive performance assessment of the solar orientation device on the solar wing, and the drive performance verification of the solar orientation device is completed.

In this paper, section 2 analyzes the system architecture, and the discrete state-space model of the studied Semi-Physical Simulation Platform is developed. Section 3 presents the design and theoretical analysis of the controller based on the terminal slide mode with an extended state observer. Section 4 shows the results that demonstrate the effectiveness of the proposed control scheme. Finally, Section 5 draws conclusions.

2 System Architecture and Dynamic Modeling

2.1 Components of Semi-Physical Simulation Platform

As shown in Fig. 1, in this paper, a set of semi-physical simulation platform of SADA is designed to simulate the variable load moment applied by the solar directional device to the solar wing during its driving process. The principle of operation is as follows: the load unit with high stiffness connects to the sun orientation device through a coupling. While the servo controller drives the sun orientation device to rotate, the rotary encoder measures the rotation angle of the sun orientation device and passes it to the dynamic simulation unit. The dynamic simulation unit calculates the load torque to be loaded on the sun orientation device based on the angle information in real time [13]. The input of the dynamic simulation unit is the angle, angular velocity, and angular acceleration during the rotation of the solar orientation device, and the output is the loading torque. The torque sensor measures the loading torque in real time and feeds it back to the drive motor for closed-loop control to improve loading accuracy. Unlike a conventional load simulator, the input to this design is the desired reaction moment, which is solved by real-time online calculations from the dynamic unit [14].

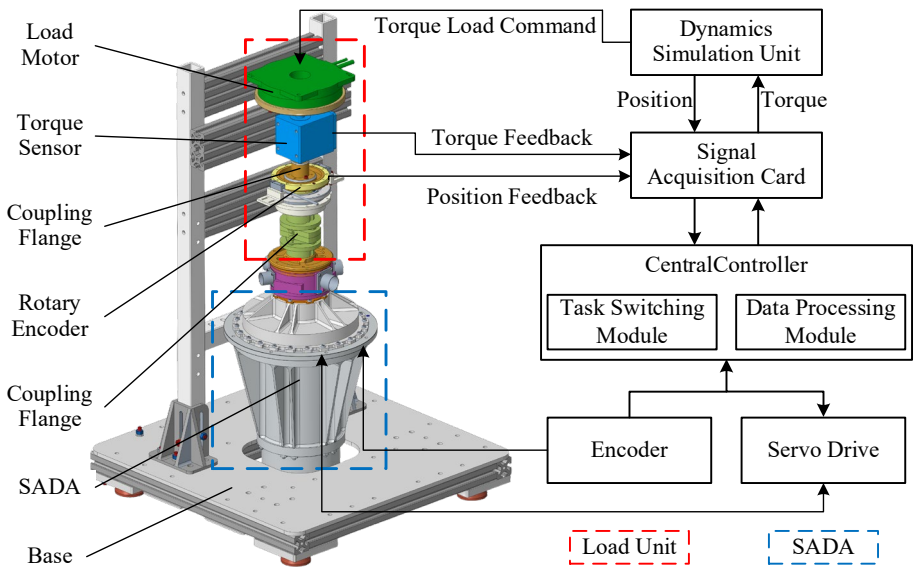


Fig. 1. Components of the semi-physical simulation platform for SADA

2.2 Mathematical Model of Load Unit

The active position motion of the SADA will strongly interfere with the closed-loop torque control of the load unit, which will generate excess torque interference in the load unit. In addition, there are many nonlinear factors in the system, such as time-varying parameters, sensor measurement noise, friction nonlinearities, etc., which seriously affect the accuracy of the loaded torque output of the load unit and also make the torque control difficult.

The loading unit uses the DC torque motor to generate the dynamic load torque applied to the SADA under test. Since the direct torque control (DTC) technique has high dynamic torque response and excellent speed control characteristics, the DTC scheme is adopted for the DC torque motor drives to obtain the load torque with low time delay, high dynamic, and high reliability [15]. In normal conditions, SADA operates within 5 Hz, while the electromagnet torque of the torque motor is updated at up to 16 KHz. Therefore, the driver of the loading motor can be modeled as an ideal electromagnetic torque generator. The dynamic characteristics of the torque motor driver can be neglected in practice. Thus, the input-output relationship of the control input $u(t)$ versus the output electromagnetic torque T_m is given by [16]:

$$T_m = K_m u(t) \tag{1}$$

where K_m is the equivalent gain of the DC torque motor driver.

According to the torque balance principle, the mechanical dynamic model of the torque motor is described as follows:

$$T_m = J_m\dot{\omega}_m + B_m\omega_m + T_l + T_f \tag{2}$$

$$\omega_m(t) = \frac{d\theta_m(t)}{dt} \tag{3}$$

where θ_m is the rotation angle, and ω_m is the angular velocity of the DC torque motor. J_m is the moment inertia realized at the load unit, B_m B_m is the viscosity coefficient, T_l represents the simulated load torque of the load unit under test and T_f represents the lumped disturbance torque caused by friction, backlash, etc.

A torque sensor mounted on the torque motor output shaft is used to measure the load torque. According to Hooke's law, the mathematical model of the torque sensor is given by:

$$T_l = K_A(\theta_m - \theta_r) \tag{4}$$

where θ_r is the angular position of SADA, K_A is the stiffness coefficient of the torque sensor. Differentiating Eq. (4), the dynamic of the torque sensor can be obtained as follows:

$$\dot{T_L} = K_A(\omega_m - \omega_r) \tag{5}$$

where ω_r denotes the angular velocity of SADA.

From the above analysis, define the load torque T_l and angular velocity ω_m as state variables, i.e., $x = [x_1, x_2]^\mathrm{T} = [T_l, \omega_m]^\mathrm{T}$, By combing Eqs. (1), (2), (4), and (5), the dynamics of the load unit can be described by the following state space.

$$\begin{cases} \dot{x_1}(t) = K_A(\omega_m - \omega_r) \\ \dot{x_2}(t) = -\frac{1}{J_m}x_1(t) - \frac{B_m}{J_m}x_2(t) + \frac{K_m}{J_m}u(t) - \frac{1}{J_m}T_f \\ y = x_1(t) \end{cases} \tag{6}$$

Eq. (6) can be rewritten as:

$$\begin{cases} \dot{x}(t) = Ax(t) + Bu(t) + f(t) + \varphi(t) \\ y(t) = Cx(t) \end{cases} \tag{7}$$

where $\boldsymbol{A} = \begin{pmatrix} 0 & K_A \\ -\frac{1}{J_m} & -\frac{B_m}{J_m} \end{pmatrix}$, $\boldsymbol{B} = \begin{pmatrix} 0 \\ \frac{K}{J_m} \end{pmatrix}$, $\boldsymbol{C} = (1 \quad 0)$, $\varphi(t)$ and $f(t)$represents lumped external disturbances and force of friction, respectively.

Give desired loading force $x_d(t) = T_d(t)$, which is assumed to be known and bounded, then its first and second derivations $\dot{x_d}$, $\ddot{x_d}$ are continuous and bounded. With unknown but bounded estimated displacement disturbance, the control objective is to design a control input $u(t)$, such that output $y = x_1$ track $x_d(t)$ as closely as possible despite the disturbance produced by various model uncertainties.

3 Slide Mode Control With ESO

3.1 Extended State Observer Design

An ESO can be applied to resolve whole modeling uncertainties in designing a system controller. $\varphi(t)$ and $f(t)$ represents lumped external disturbances and force of friction, respectively, and are treated as generalized disturbances, which denote the deviation from the nominal system model. Eq. (7) can be rewritten as:

$$\begin{cases} \dot{x}(t) = \boldsymbol{A}x(t) + \boldsymbol{B}u(t) + d \\ y(t) = \boldsymbol{C}x(t) \end{cases} \tag{8}$$

where $d = \varphi(t) + f(t)$, extending d as an additional state variable, i.e., define $x_3 = d$, in which x_3 is an extended system state, and let $h(t)$ represent the time derivative of d, the original plant in Eq. (6) can be described as

$$\begin{cases} \dot{x}_1(t) = K_A(\omega_m - \omega_r) \\ \dot{x}_2(t) = -\frac{1}{J_m}x_1(t) - \frac{B_m}{J_m}x_2(t) + \frac{K_m}{J_m}u(t) + x_3 \\ \dot{x}_3(t) = h(t) \end{cases} \tag{9}$$

The loading torque output from the load unit and the rotational speed of the motor can be easily measured by the torque sensor and the angle sensor, i.e., the state variables x_1, x_2 can be obtained in real time. In order to facilitate the design of a suitable controller to realize the accurate tracking of the desired loading moment $x_d(t)$ by load unit in the presence of many disturbances in the system, the following assumptions are given [17]:

- **Assumption 1**: $x_d(t)$ and $\theta_r(t)$ are continuous, and its first and second derivations are continuous and bounded.
- **Assumption 2**: T_f is continuous, and its derivations are continuous and bounded.
- **Assumption 3**:d is continuously bounded and satisfies the Lipschitz condition.

To estimate the value of x_1, the following ESO is designed [18]:

$$\begin{cases} \dot{\hat{x}}_1 = K_A\hat{x}_2 + \lambda - \beta_1(\hat{x}_1 - x_1) \\ \dot{\hat{x}}_2 = f(x_1, x_2) + \frac{K_m}{J_m}u + \hat{x}_3 - \beta_2(\hat{x}_1 - x_1) \\ \dot{\hat{x}}_3 = -\beta_3(\hat{x}_1 - x_1) \end{cases} \tag{10}$$

where $\beta_1, \beta_2, \beta_3$ are gain coefficients of ESO, $\dot{\hat{x}}_1, \dot{\hat{x}}_2, \dot{\hat{x}}_3$ are estimates of the state variable x_1, x_2, x_3, let $\tilde{x}_i = x_i - \hat{x}_i$, $i = 1,2,3$, denote the estimation error. From Eqs. (9) and (10), the observer estimation error can be shown as

$$\begin{cases} \dot{\tilde{x}}_1 = K_A\tilde{x}_2 - \beta_1\tilde{x}_1 \\ \dot{\tilde{x}}_2 = \tilde{x}_3 - \beta_2\tilde{x}_1 \\ \dot{\tilde{x}}_3 = -\beta_3\tilde{x}_1 - h(t) \end{cases} \tag{11}$$

associative Eqs. (9) and (11), we can get

$$\dot{\tilde{x}} = \boldsymbol{A}\tilde{x} + \boldsymbol{M}h \tag{12}$$

where $\dot{\tilde{x}} = (\dot{\tilde{x}}_1 \quad \dot{\tilde{x}}_2 \quad \dot{\tilde{x}}_3)^T$, $\tilde{x} = (\tilde{x}_1 \quad \tilde{x}_2 \quad \tilde{x}_3)^T$,$\boldsymbol{A} = \begin{pmatrix} -\beta_1 & K_A & 0 \\ -\beta_2 & 0 & 1 \\ -\beta_3 & 0 & 0 \end{pmatrix}$, $\boldsymbol{M} = \begin{pmatrix} 0 \\ 0 \\ -1 \end{pmatrix}$.
To simplify the parameter design, the eigenroots of matrix $\boldsymbol{A}$ are configured at the closed-loop bandwidth of the system using the pole configuration method and the computed parameters [19], which can make matrix $\boldsymbol{A}$ conform to the Hurwitz matrix condition.

$$\lambda(s) = |s\boldsymbol{I} - \boldsymbol{A}| = (s + \omega_o)^3 \tag{13}$$

Expanding Eq. (13) yields the gain coefficients $\beta_1, \beta_2, \beta_3$, which are

$$\begin{cases} \beta_1 = 3\omega_0 \\ \beta_2 = \frac{3}{K_A}{\omega_0}^2 \\ \beta_3 = \frac{{\omega_0}^3}{K_A} \end{cases} \tag{14}$$

From the dynamic of the scaled estimation errors in Eq. (14), Lyapunov equation, by borrowing the stabilizing method utilized in high-gain observers and based on primary analysis, it can be inferred that the designed ESO in Eq. (11) is stable and the state estimation error can be made arbitrarily small by increasing the bandwidth ω_0.

3.2 Slide Mode Control Design

According to the sliding mode control method, a sliding mode controller is designed for Eq. (10). The linear sliding surface is the following:

$$s = (\dot{x}_1 - \dot{x}_d) + c(x_1 - x_d) \tag{15}$$

The sliding mold surface convergence law is chosen to be an exponential convergence law, as shown in the following equation:

$$\dot{s} = -ks \tag{16}$$

So, the slipping surface s is reachable. By combining Eqs. (10), (15), (16),

$$\begin{aligned} \dot{s} &= (\ddot{x}_1 - \ddot{x}_d) + c(\dot{x}_1 - \dot{x}_d) \\ &= K_A(\dot{x}_2 - \dot{\omega}_r) - \ddot{x}_d + c[K_A(x_2 - \omega_r) - \dot{x}_d] \\ &= K_A\left(-\frac{1}{J_m}x_1 - \frac{B_m}{J_m}x_2 + \frac{K_m}{J_m}u - d - \dot{\omega}_r\right) - \ddot{x}_d + cK_A(\dot{x}_2 - \dot{\omega}_r) - c\dot{x}_d \\ &= -ks \end{aligned} \tag{17}$$

The sliding mode control law u of the load unit is computed as follows:

$$u = \frac{J_m}{K_A K_m}[\ddot{x_d} + c\dot{x_d} - k(\dot{x_1} - \dot{x_d}) + kc(x_1 - x_d)]$$
$$+ \frac{J_m}{K_m}(\dot{\omega_r} - cx_2 - c\omega_r) + \frac{1}{K_m}(x_1 + B_m x_2 + d) \tag{18}$$

To verify the validity of the above design, the study hereby gives proof of the above slip mold surface based on the Lyapunov stability theory [20]. The following Lyapunov function is selected:

$$V = \frac{1}{2}s^2 \tag{19}$$

where $c > 0$, $k > 0$, then, $V = S\dot{S} = -kS^2$, when $S \neq 0, S\dot{S} < 0$. As a result, this linear sliding mold control is stable, and the system can reach the sliding surface from the initial state regardless of where that state is located.

Combining the ESO and the linear sliding mode controller, the servo block diagram of the system is shown in Fig. 2.

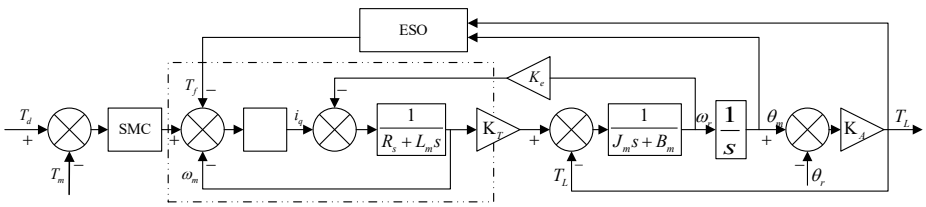


Fig. 2. Loading unit control block diagram

In the figure:T_d is commanded torque, SMC is a sliding mode controller, and DTC is a DC Torque Motor Driver. When the main control computer gives the command torque, the load unit will correspondingly output a certain moment. ESO is used to estimate and compensate for the unknown term perturbation generated by the loading unit in the execution process; the sliding mode is used as the torque loop controller, which helps to improve the response speed of the loading motor and the loading accuracy.

4 Mathematical Simulation

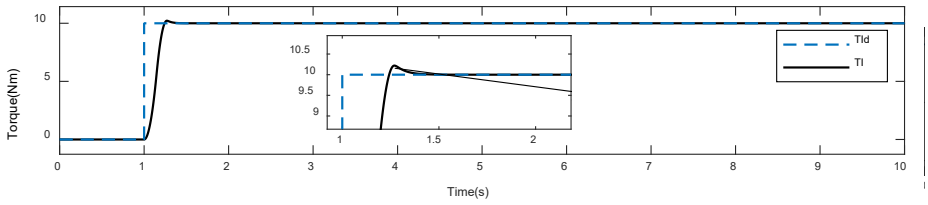
In order to verify the feasibility of the research method, the moment following simulation experiments are carried out for step input signal, sinusoidal input signal, and solar wing torsional vibration signal, respectively. Different interference signals are input to analyze the anti-interference capability of the system. In the simulation experiments:

The parameters of the linear slip film controller are selected as follows: $K = 50, c = 1$; the physical parameters of the DC torque motor are shown in Table 1.

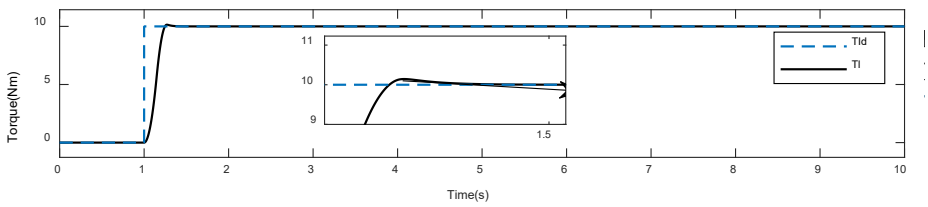
Table 1. Torque Motor Physical Parameters

Parameter	Value	Parameter	Value
K_A	8000 N·m/rad	R_m	20.1 Ω
J_m	800 kg·m^2	L_m	0.83 H
B_m	0.008	K_e	5.73 V/rad/s
K_m	1.317 N·m/A	K_p	2.4

Fig. 3 (a) and (b) show the moment response curves of the loading unit to a step signal when a step signal perturbation is applied at 2 seconds and a sinusoidal perturbation signal at 1 Hz, respectively. It can be seen that under the step and sinusoidal perturbation, the torque output signal only has a very small overshoot in a short time, and the response time is also very short, which has a strong anti-interference ability.



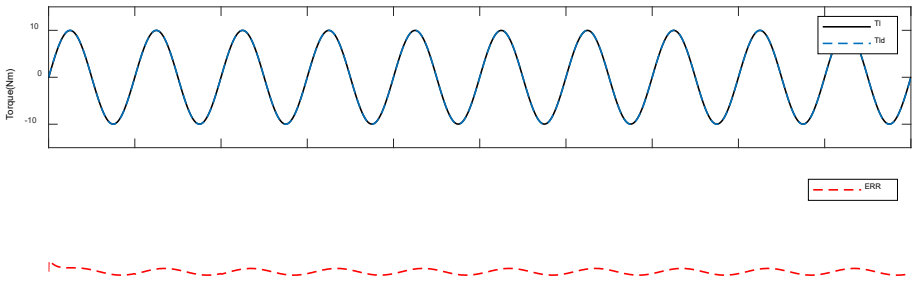
(a)



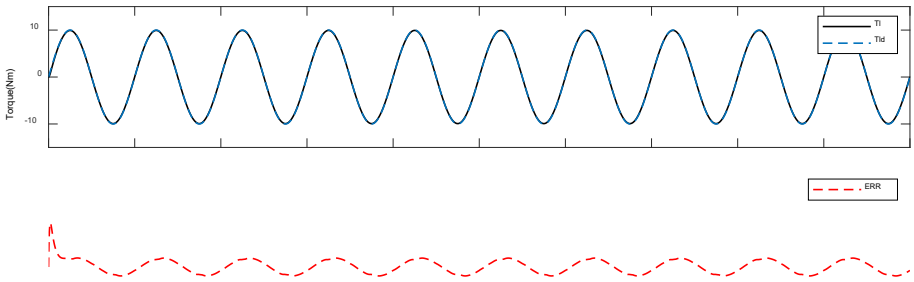
(b)

Fig. 3. Step-signal Moment Response under Step and Sinusoidal Perturbation

Fig. 4 (a), (b), (c) shows the moment response curves of the loading unit to 1 Hz, 5 Hz, and 10 Hz sinusoidal signal inputs for 1 Hz sinusoidal perturbations, respectively. It can be seen that under sinusoidal perturbation, when the frequency of the input signal is larger, the error of the system in the initial adjustment is larger, but within 0~5 Hz of the system operation, the error in normal operation is less than 0.5 Nm, and the response time of adjustment is about 2 ms, which is capable of adjusting the system to be within the tolerance range within a relatively fast time.



(a)



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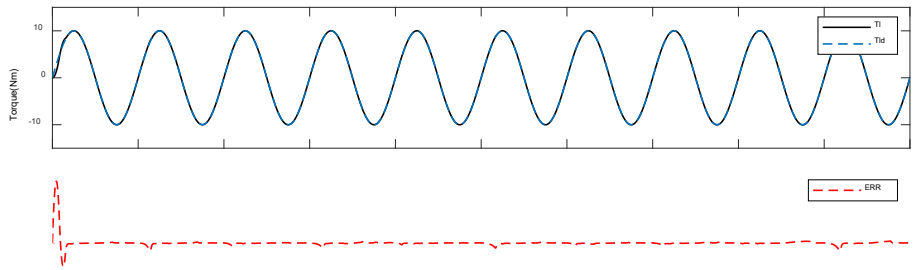


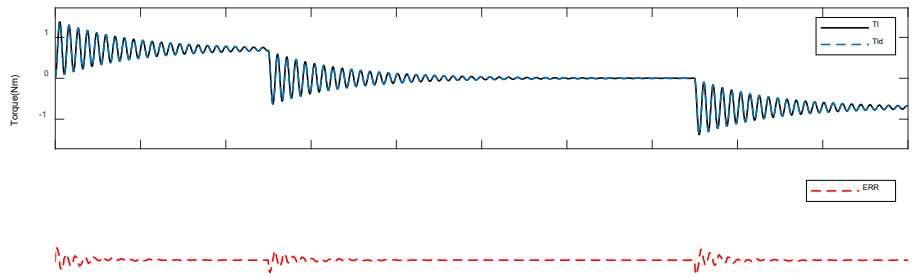
(c)

Fig. 4. Sinusoidal Moment Response under Sinusoidal Perturbation

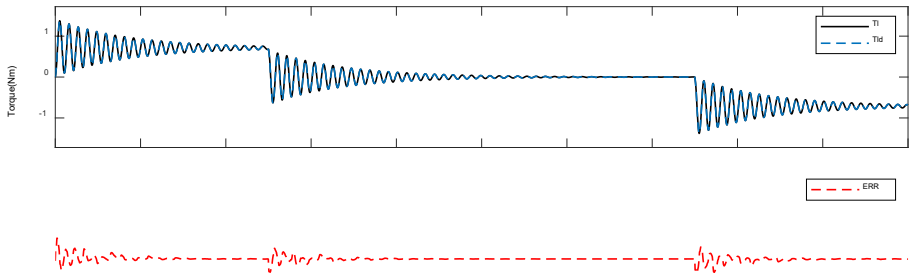
In the actual on-orbit environment, the reaction moment of the solar wing is a torsion-vibration signal in which the torsion and vibration signals are superimposed on each other [21].

Fig. 5 (a) and (b) show the moment response curves of the loading unit to the torsional vibration signal when a step signal perturbation is applied at 2 seconds and a sinusoidal perturbation signal at 1 Hz, respectively.

As shown in Fig. 5, when the input is the torsional vibration reaction moment generated by the actual on-rail, under the perturbation of the step and sinusoidal signals, the moment-following error is nearly 0, and the difference in amplitude is below 1%, which has a good moment tracking performance.



(a)



(b)

Fig. 5. Torsion-Vibration Signal Moment Response under Step and Sinusoidal Perturbation

5 Conclusion

Based on an analysis of the influences on the accuracy of the loading unit, an ESO is designed to estimate the internal errors and external disturbances present in the system, and a torque loop controller is designed using linear sliding film control. Through the simulation verification of the step signal, sinusoidal signal, and actual on-rail torsional-vibration signal, it is shown that the system has a very good performance in resisting external perturbation and torque following.

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